1.How to define objects in JavaScript. Explain with Example.

An **object** in JavaScript is a collection of **key-value pairs**. It can store properties (data) and methods (functions).

The most common way to create an object is using **object literal syntax**:

```
let student = {
  name: "Akanksha",
  age: 22,
  course: "Full Stack Development",

  greet: function() {
    console.log("Hello!");
  }
};
```

Here:

- **name**, **age**, and **course** are **properties**.
- **greet()** is a **method**.

Another way is using **new Object()**:

```
let student = new Object();
student.name = "Akanksha";
student.age = 22;
```

2.How do you access properties and methods of JavaScript objects. Explain with example.

There are **two common ways** to access object properties:

A. Dot (.) notation

```
let student = {
  name: "Akanksha",
  age: 22
```

```
};


console.log(student.name);

console.log(student.age);
```

Output:

Akanksha

22

To call a method:

```
let student = {
   name: "Akanksha",


   greet: function() {
      console.log("Hello " + this.name);
   }
};
student.greet();
```

Output:

Hello Akanksha

B. Bracket ([]) notation

```
console.log(student["name"]);

console.log(student["age"]);
```

Bracket notation is especially useful when the property name is dynamic:

```
let property = "name";

console.log(student[property]);
```

Output:

Akanksha

3.What is the use of this keyword in JavaScript?

The **this keyword refers to the object associated with the current function call**. In an object method, this commonly refers to the object that called the method.

Example:

```
let student = {
   name: "Akanksha",
   age: 22,

   display: function() {
      console.log(this.name);
      console.log(this.age);
   }
};

student.display();
```

Output:

Akanksha

22

Here, **this.name** means **student.name** and **this.age** means **student.age**.

In simple words:

this is used to refer to the current object and access its properties or methods.